Data Cleaning & Preparation Report

Project: Mobile Money Transaction Pattern Analysis — CSC 3221

1. Data Quality Issues Found

Issue	Variable	Count	Percentage	Action Taken
Missing values	amount_xaf	43	1.46%	Imputed with median by tx type
Missing values	transaction_type	30	1.02%	Imputed with mode
Duplicate rows	All columns	8	0.27%	Removed
Negative amounts	amount_xaf	5	0.17%	Absolute value applied
IQR outliers	amount_xaf	169	5.8%	Winsorized at IQR fences
Failed transactions	status	138	4.7%	Excluded from feature eng.

2. Cleaning Operations Applied

- **Duplicate removal:** 8 exact duplicate transaction rows removed using pandas drop_duplicates().
- **Negative amount correction:** 5 transactions had negative amounts (likely sign errors in data entry). Corrected by taking absolute values.
- **Missing amount imputation:** 43 missing transaction amounts imputed using the median amount for each transaction type (grouped median imputation), preserving type-specific value distributions.
- **Missing transaction type imputation:** 30 missing transaction types imputed with the most frequent class (Send Money).
- **Outlier treatment (IQR method):** IQR fences computed as Q1 - 1.5*IQR and Q3 + 1.5*IQR. Amounts below the lower fence or above the upper fence were winsorized (capped at fences) rather than removed, preserving sample size.
- **Date standardization:** All date fields converted to pandas datetime format. Day of week, month, quarter, and year columns extracted.
- **Status filtering:** Only 'Success' transactions used for feature engineering (failed/pending excluded from behavioral metrics).

3. Feature Engineering

Feature	Description	Rationale
tx_per_month	Transactions per month	Core activity frequency metric
avg_transaction_amount	Mean transaction value	Financial scale indicator
send_receive_ratio	Sent / Received ratio	Behavior directionality
weekend_ratio	Weekend transactions / total	Lifestyle pattern proxy
days_since_last_tx	Days since most recent tx	Recency (RFM model)
avg_monthly_amount	avg_amount * tx_per_month	Total financial volume
tx_type_diversity	Unique transaction types used	Service adoption breadth
total_fees_paid	Sum of all fees	Proxy for total volume

active_months	Duration of activity period	Tenure indicator

4. Summary Statistics Before and After Cleaning

Statistic	Before (amount_xaf)	After (amount_xaf)
Count	2,895	2,930
Mean	38,645 XAF	35,990 XAF
Std Dev	41,711 XAF	32,516 XAF
Min	-125,100 XAF	100 XAF
25th Percentile	8,450 XAF	8,900 XAF
Median	27,000 XAF	27,100 XAF
75th Percentile	52,950 XAF	52,100 XAF
Max	297,800 XAF	116,900 XAF

After cleaning, the distribution is more representative with no extreme negative values, reduced variance (std dropped from 41,711 to 32,516 XAF), and a more conservative max value following outlier treatment. The cleaning process improved data quality without significantly altering the central tendency.

5. Data Transformation for Modeling

- **Label encoding (ordinal):** education_level (0-4), monthly_income_range (0-4), zone_type (0-2), activity_segment (0-2 as target).
- **Binary encoding:** gender (Male=1), smartphone_owner (Yes=1).
- **One-hot encoding:** primary_mm_use, mm_provider (nominal variables with no natural order).
- **Standard scaling:** Applied to all features before Logistic Regression and KNN to ensure equal feature contribution.
- **Train/Val/Test split:** 70% / 15% / 15% stratified split ensuring class balance across splits.